

Po-Chun Wu

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Education

Carnegie Mellon University

Master of Science in Electrical and Computer Engineering

Master of Science in Engineering and Technology Innovation Management (Dual Degree)

Relevant Courses: Fundamentals of Computer Systems, Optimization, Networks in the Real World

Pittsburgh, PA

May 2028

May 2028

National Taiwan University

Bachelor of Science in Electrical Engineering

GPA: 4.16/4.3

Relevant Courses: Data Structures, Algorithms, Machine Learning, SoC Verification, Computer Architecture

Taipei, Taiwan

Jan 2026

Projects

Formal Verification – NTU, Research Project (Advising: Prof. Chung-Yang Ric Huang, Team of 2) Apr 2025 — Jun 2025

- Modeled optimal vocabulary learning sequence generation as an NP-hard Target Set Selection variant.
- Implemented bounded model checking-based verification to derive optimal solutions for graphs with 150 nodes.

LifeOS Automation System – NTU, Course Project (Team of 6) Jan 2025 – Mar 2025

- Designed a backend Notion plugin to synchronize entries across different databases.
- Engineered a timestamp-based version-state model to resolve sync conflicts between Notion pages and MongoDB records.
- Developed a document-linking layer mapping a canonical schedule to multiple nodes for one-to-many synchronization.

AR Navigation Helmet – Best Maker Award, 2024 MakeNTU Competition (Team of 5) May 2024

- Created a navigation assistant mounted inside a motorcycle helmet, using speech recognition and AR to help riders navigate.
- Devised a multi-threaded system in Python to handle asynchronous voice input, API requests, and OLED rendering.

NTU Pair – Personal Project Dec 2022 – Feb 2023

- Built a full-stack dating app through *vanilla JS*, reaching 1,600+ verified users from NTU.
- Formulated a matchmaking and recommendation algorithm to match users based on psychological traits.
- Re-architected the system with secure coding practices after mitigating XSS, DoS, and credential-stuffing attacks.

Work Experience

SLEKMED Taipei, Taiwan (Remote)

Product Developer (Part-Time)

Aug 2023 – Mar 2026

- Note-to-Video Generator:** Developed a web-based note-to-video generator using Marp, OpenAI API, ElevenLabs API, and FFmpeg, enabling medical experts to create high-quality educational videos from Markdown notes in less than 10 minutes.
- Medical Simulation Platform:** Created a Next.js-based simulation module for medical diagnostic training in collaboration with Taipei Medical University peers, gaining 4,200+ followers and securing funding from the Taiwan Ministry of Education.
- Virtual Patient Chatbot:** Implemented an LLM-based chatbot as a virtual patient, contributing to a 1st-place finish among 40+ teams in the 2024 CTCL AI Competition (12K USD prize).

Skills

Programming: Python, JavaScript/TypeScript, C++, Verilog

Frameworks/Tools: React/Next.js, Node.js, Asyncio, MongoDB, MiniSat, Git

Specialties: System design, algorithm design, full-stack development, asynchronous programming, automated reasoning

Leadership

Founder and President, NTU Learning Optimization Club Aug 2023 – Jun 2025

- Researched learning methodologies, delivered 8 nationwide speeches, and advised 200+ students on study habits.

Makerspace Administrator, NTUEE Student Association Academic Department Sep 2023 – Dec 2023

- Organized 25 academic workshops and taught microcontroller programming to electrical engineering undergraduates.

Project Lead, NTUEE Virtual Makerspace May 2023 – Aug 2023

- Led a 25-member team to build a React/Node.js inventory and reservation system for the university student makerspace.